



Answers & Solutions:

1. **B. band to band transition dominants over impurity ionization**
2. **A. the anode voltage must be positive with respect to its cathode.**
3. **A. towards positive terminal**
4. **D. Both A and C.**
5. **D. Voltage regulator**
6. **D. Germanium**
7. **A. ± 4 .**
8. **A. Both A and R are true and R is correct explanation of A**
9. **B. 7.19 mA**

Solution:

$$I_D = I_s \left(e^{\frac{kV_D}{T_k}} - 1 \right)$$

$$I_D = (50 \text{ nA}) \left(e^{\frac{\left(\frac{11600}{2}\right)(0.6)}{(20+273)}} - 1 \right)$$

$$I_D = 7.19 \text{ mA} \rightarrow \text{Ans.}$$

where:

$$k = \frac{11600}{n}$$

$n = 1$ for Ge

$n = 2$ for Si

10. **C. It is expensive to produce**
11. **D. holes and electrons**
12. **C. The reverse current**
13. **B. Both A and R are true but R is not a correct explanation of A**
14. **C. The junction will conduct current**
15. **A. Reverse biased**

16. **B. Is positive relative to the cathode**
17. **C. covalent bonding.**
18. **A. Minimize the diode capacitance**
19. **A. 175 nA**

Solution:

$$I_{s1} = I_{s0} e^{k(T_1 - T_0)}$$

$$I_{s1} = 35 \text{ nA} e^{0.07(50-27)}$$

$$I_{s1} = 175 \text{ nA} \rightarrow \text{Ans.}$$

where:

$$k = 0.07 / ^\circ \text{C}$$

20. **B. closed switch with a voltage drop of 0 V.**
21. **B. Reverse voltage**
22. **A. Reverse bias**
23. **C. Avalanche voltage**
24. **C. pentavalent impurity atoms.**
25. **C. Give a semiconductor material certain properties**
26. **D. The substance has an electron surplus**
27. **B. intrinsic semiconductor.**
28. **D. None of the above**
29. **A. The frequency**
30. **C. A charge of +1 unit**
31. **B. Charge carriers are passed from atom to atom**
32. **D. holes.**
33. **B. reverse-biased.**
34. **A. silicon**
35. **C. Requires no battery**
36. **B. About 0.6 V**

37. **D. Ability to withstand high voltages.**
 38. **B. 3**
 39. **C. An IMPATT diode**
 40. **A. trivalent impurity atoms.**
 41. **B. Is employed in some radio receivers**
 42. **A. Photocells**
 43. **B. Adding a donor impurity**
 44. **B. $1.6 \times 10^{-19} \text{ C}$**
 45. **B. Plus to minus**
 46. **D. Can cause objectionable signal distortion**
 47. **D. holes.**
 48. **D. Can cause objectionable signal distortion**
 49. **B. 0.053 %/ $^\circ \text{C}$**
- Solution:
- $$V_f = V_i (1 + k\Delta T)$$
- $$5 \text{ V} = 4.8 \text{ V} [1 + k(25-100)]$$
- $$k = 0.053\% / ^\circ \text{C}$$
50. **B. Variable conductive properties of some materials.**

END